

Task 3: One-Page Design Summary — Jurisdiction-Aware Compliance Layer

1. Core Architectural Design and Data Pipeline

The **CredVeritas Compliance Engine** is engineered as a decoupled, independent compliance auditing layer designed to reconcile the cross-border regulatory frictions faced by DBS Bank Ltd. when deploying automated credit underwriting models across overlapping jurisdictions. In contemporary digital lending practicalities, financial institutions frequently utilize identical core machine learning models globally to optimize computational efficiency. However, this creates an operational compliance crisis due to the political and philosophical schisms between regulators such as the US CFPB and the Monetary Authority of Singapore (MAS).

To resolve this without degrading core credit decision workflows, CredVeritas implements a strategic architectural choice: **the isolation of sensitive attributes via a "Shadow Labels" framework**. During the initial model training phase (utilizing `training_data.csv` comprising 1,120 historical baseline credit profiles), demographic features such as Sex and Age are strictly stripped from the feature matrix to eliminate direct discrimination from the underwriting engine from the outset. Crucially, instead of discarding these variables entirely, CredVeritas securely sequesters them within an independent, isolated compliance auditing database layer. When live production transactional streams enter the pipeline (simulated via the closed-loop ingestion of 72 records compiled by merging `test_data.csv` and `kaggle_sample_solution_file.csv`), the engine's jurisdiction-aware gateway intercepts the payloads and routes them dynamically based on geographic network origin (Geo-IP), enforcing distinct jurisdictional compliance trees on a single credit model base.

2. Jurisdictional Routing Mechanics

- **US Jurisdiction Mode (Micro-Level Individual Explainability):** When the gateway routes transactions under the US rule tree, the engine aligns with the statutory mandates of CFPB Regulation B and the Equal Credit Opportunity Act (ECOA). Full-stream scanning of the 72 live production records identifies 34 loan denial decisions (Risk == bad). The engine immediately executes an explainability pipeline powered by model-agnostic feature attribution. By computing the structural mathematical contribution of specific credit variables, the system bypasses the algorithmic "black box" to isolate the principal reasons for credit rejection for every unique consumer. For example, for Applicant ID 240, the engine dynamically isolates a mismatch where the requested credit amount (\$4,473) is excessive relative to a prolonged repayment duration (36 months). For Applicant ID 919, the primary friction is diagnosed as an insufficient status in the primary checking account. Within milliseconds, these quantitative attributions are compiled into compliant, natural-language, human-

readable **Adverse Action Notices**, fulfilling the rigid individual transparency mandates of US consumer law.

- **Singapore Jurisdiction Mode (Macro-Level Systemic Parity):** When toggled to the Singapore rule tree, the engine automatically aligns with the MAS FEAT Principles and the 2024 AI Model Risk Management (MRM) guidance. The system seamlessly suppresses individual-level explanations and shifts its analytical aperture to portfolio-wide systemic fairness. Utilizing the secured "Shadow Labels," the engine tracks conditional acceptance distributions across protected cohorts to calculate the continuous **Disparate Impact (DI) Ratio**. Empirical scanning of the baseline live stream reveals a critical metric breach: the system registers a baseline DI ratio of **1.3181**, significantly exceeding the upper prudential 合规 boundary of **1.25** (the four-fifths rule). This mathematically demonstrates that the core credit model is executing a silent proxy discrimination, resulting in a systemic over-rejection and "de-banking" of the male demographic cohort. The engine immediately flags a **Critical Compliance Violation**, halts non-compliant pipeline tranches, and routes an automated enforcement alert to the Chief Compliance Officer (CCO).

3. Quantitative Resilience via Stress Testing (Option C Alignment)

CredVeritas fundamentally rejects superficial, checklist-based compliance. To fulfill the advanced quantitative requirements of Option C, the engine embeds an automated **distortion stress testing module** to evaluate threshold resilience under macroeconomic shocks and data drift.

In the simulation, a severe macroeconomic contraction disproportionately affecting specific borrower segments was injected into the pipeline, introducing a +20% risk perturbation on the female demographic cohort. Upon running the stressed production stream, the metric shifted counter-intuitively: the Disparate Impact ratio compressed down from its non-compliant baseline to a compliant level of **1.0784** (in [0.80, 1.25]). This finding delivers a profound, non-trivial insight to senior management: **static, retroactive compliance reporting is highly hazardous; a model that appears perfectly compliant on a static year-end summary (DI = 1.0784) may actually be a severely drifted model that underwent a volatile macro-distortion from an initially non-compliant state (DI = 1.3181)**. This empirical validation confirms the core thesis of CredVeritas: continuous, real-time transactional stream scanning coupled with quantitative sensitivity analysis is the only technical mechanism capable of capturing dynamic compliance inversions and mitigating genuine institutional risk.